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RESEARCH ARTICLE

YOLOv8n-ASF-DH: An Enhanced Safety Helmet Detection Method

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ABSTRACT Wearing a safety helmet is an essential aspect of safety production management. To address the issue of low detection accuracy of existing safety helmet detection algorithms for small targets and complex scenes, we propose the YOLOv8n-ASF-DH model (DH stands for Dynamic Head), which is based on YOLOv8n and integrates the Attentional Scale Sequence Fusion (ASF) structure and the Dynamic Head (DyHead). In the backbone layer, a Triplet Attention mechanism is incorporated to enhance the model's focus on features of small targets. In the neck layer, the ASF structure efficiently fuses different level output features extracted by the backbone network, enhancing the overall feature representation capability of the model. In the detection head, DyHead adjusts the relationships between different feature layers, enhancing the model's performance on different scales and complex scenes. Additionally, adopting the Focal-EIoU loss function balances the contributions of high-quality and low-quality samples in loss calculation. Experimental results demonstrate that the improved model enhances detection performance in complex scenes and small target scenarios. Compared to the YOLOv8n model, the YOLOv8n-ASF-DH model achieved an improvement of 3.066% in accuracy and 3.883% in recall. Additionally, the model exhibited an increase of 2.584% in mAP 0.5 and 6.131% in mAP 0.5:0.95. Compared with other mainstream object detection algorithms, the improved model significantly enhances the performance of safety helmet target detection tasks.

INDEX TERMS Helmet wearing detection, YOLOv8, attentional scale sequence fusion, dynamic head, triplet attention mechanism.

I. INTRODUCTION

The construction site environment for high-altitude operations is complex and intricate, with a constant risk of falling objects [1]. Therefore, the requirement for workers on construction sites to wear safety helmets is paramount to minimizing the risk of head injuries and protecting life safety [2].

Compared to the inefficient and slow process of manually checking each worker's compliance with safety helmet requirements through monitors, these methods are labor-intensive and prone to errors, necessitating significant effort to oversee extensive areas. Therefore, the development of high-performance safety helmet detection algorithms is necessary. Numerous scholars have researched safety helmet detection, predominantly utilizing deep learning methods. The performance of safety helmet detection has been significantly improved through computer vision algorithms. Due to its excellent detection performance, deep learning has been widely applied in safety helmet recognition, offering a superior solution to traditional methods [3].

The current safety helmet recognition algorithms mainly include one-stage and two-stage detection methods. One-stage object detection methods complete object location estimation and category classification in a single computation, offering faster speeds but potentially higher false-positive rates in detecting small objects. On the other hand, two-stage methods are realized through two independent network

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models, achieving higher precision, especially in large objects and complex scenes, albeit at slower speeds. Although two-stage detection processes are precise, their computational intensity and slowness render them unsuitable for real-time applications. Conversely, one-stage detection methods simplify the process by integrating region proposal and object classification, providing a faster solution.

Among one-stage detection methods, the YOLO (You Only Look Once) series algorithms have balanced precision and processing speed [7]. Despite these advancements, YOLO-based algorithms still face challenges detecting small objects against complex backgrounds when detecting safety helmets. Particularly, backgrounds comprising buildings and vegetation may affect the detection and recognition of safety helmets. Moreover, the relatively small size and uniform color further exacerbate their susceptibility to misidentification in complex environments. Additionally, the detection of safety helmets also presents occlusion issues, where helmets may be partially or entirely obscured by the background, leading to the algorithm's inability to recognize the target accurately.

Building upon the foundation of YOLOv5, the YOLOv8 algorithm introduces novel improvements that showcase exceptional performance in object detection, striking an unprecedented balance between accuracy and speed [4], [5]. Although enhancements in the YOLO algorithm for safety helmet detection have been made, single-stage algorithms may still need to improve accuracy when detecting small objects or in the presence of complex background interference. Thus, there is an urgent need for algorithmic improvements to achieve superior performance in complex environments.

This research will utilize the Safety-Helmet-Wearing-Dataset (SHWD) [31] with the YOLOv8n-ASF-DH network to detect helmet wear. The goal is to bridge the gap in the helmet detection literature and provide a robust, automatic detection method. The YOLOv8n-ASF-DH model enhances the YOLOv8n framework, integrating multiple improvement modules. These enhancements boost the model's accuracy and efficiency in addressing specific challenges, such as small objects or complex backgrounds, by introducing the following key modules to improve network performance and enhance the detection capability for safety helmet targets:

1) Integration of the Triplet Attention mechanism in the Backbone Layer: By incorporating the Triplet Attention mechanism in the Backbone Layer, the YOLOv8n-ASF-DH model optimizes the feature extraction stage. This mechanism enhances the model's focus on densely packed small objects by more accurately concentrating on important feature regions within the image, significantly improving detection precision for such small targets. This mechanism is particularly suited for recognizing small, critical objects like safety helmets.
2) Application of the Attentional Scale Sequence Fusion (ASF) Structure in the Neck Layer: Introducing the ASF structure further strengthens the model's ability to recognize small object details. By merging spatial and multi-scale features, the ASF structure efficiently fuses output features from different levels (P3, P4, and P5) extracted by the backbone network. This efficient feature fusion strategy enhances the model's detection capability for small objects and improves the model's overall feature representation ability.
3) Use of DyHead in the Detection Head: Employing DyHead facilitates the synergistic effect of multiple self-attention mechanisms across feature layers, effectively promoting a comprehensive enhancement of the model's scale awareness, spatial awareness, and task awareness. DyHead enhances the model's performance across different scales and complex scenarios by adjusting the relationships between different feature layers, making the model more sensitive to detail variations and thus improving the detection accuracy for small objects.
4) Adoption of the Focal-EIoU Loss Function: By employing the Focal-EIoU Loss Function, the model achieves a balanced adjustment in the contribution of high-quality and low-quality samples during loss calculation. This method enhances the model's focus on high-quality samples (with larger IoU values) by increasing their weight while reducing the impact of low-quality samples (with smaller IoU values), effectively optimizing the learning process and improving model performance.

Through these advanced improvements and optimization designs, the YOLOv8n-ASF-DH model significantly enhances its detection performance, especially in detecting miniature target safety helmets. The improved model provides strong technical support for efficient and accurate target detection in complex environments.

This research is significant because it has the potential to fill a critical gap in the literature by offering an advanced, automatic method for detecting helmet-wearing. By leveraging the capabilities of the YOLOv8n-ASF-DH network, this study aims to contribute significantly to the field of safety and health monitoring, potentially reducing the incidence of head injuries in traffic and workplace environments.

The remainder of this document is organized as follows: Section II provides a detailed review of the existing literature. Section III elaborates on the architecture of the proposed model. Section IV details the experimental analysis results and engages in a discussion. Section V confirms YOLOv8n-ASF-DH's high efficiency, accuracy, and robustness, especially in complex scenarios, making it suitable for real-world applications. Finally, Section VI offers a summary and conclusion of the research presented.

II. LITERATURE REVIEW

In safety production monitoring at construction sites, the detection of safety helmets constitutes a foundational task, providing a quintessential application scenario for corporate intelligent video surveillance systems. In recent years,

research on safety helmet detection has increased, with methodologies broadly categorized into two groups: those based on traditional machine learning and those based on deep learning approaches.

Despite certain limitations inherent in machine learning-based detection methods, notable progress has been achieved. Specifically, Wu et al. [6] introduced a color-based hybrid descriptor, incorporating local binary patterns, Hu moment invariants, and color histograms for the extraction of features from helmets of various colors. Subsequently, a hierarchical support vector machine was constructed to classify all features, achieving an average accuracy rate of 90.3%. Additionally, a detection method based on histograms of oriented gradients (HOG) and support vector machines (SVM) was proposed to extract safety helmet features more effectively. For instance, Rubaiyat et al. [7] combined frequency domain information of images with human detection algorithms to detect construction workers and then applied color-based and circular Hough transform-based feature extraction techniques for helmet detection. Jin et al. [8] initially used a deformable part model to locate workers and then extracted potential helmet regions through color feature matching. Finally, safety helmets were detected using an SVM trained with HOG features. A method combining a latent support vector machine (latent SVM), sensitive to challenging negative examples in data mining, was employed to complete data training [9].

With the rapid advancement of deep learning in recent years, numerous methods based on convolutional neural networks (CNNs) have also achieved commendable results in safety helmet detection. Deep learning-based helmet detection technologies can be divided into two categories: two-stage detection algorithms based on candidate regions and one-stage regression-based detection algorithms. Although two-stage object detection methods offer higher accuracy, they typically require more inference time, which does not meet the needs for real-time detection [10], [11].

These single-stage deep learning-based safety helmet detection algorithms, with their straightforward structure and rapid detection capabilities, have become the preferred solution for helmet detection tasks. However, Single-stage methods remain widely favored despite a slight reduction in detection accuracy compared to the more complex two-stage helmet detection algorithms. Many scholars have proposed improvements to address the issue of lower detection accuracy in single-stage methods. For example, researchers in [12] introduced additional detection layers on YOLOv5 and used more detection heads to detect targets of different scales, enhancing the model's detection capability in complex scenarios. The model also incorporated a bidirectional feature pyramid network (BiFPN). Adding skip connections better integrates shallow semantic features and significantly reduces the model's error rate. Kun et al. [13] added a fourth detection scale in YOLOv5 to detect smaller objects and introduced an attention mechanism in the backbone network to construct more informative features. Another study [14] proposed a novel object detection model based on an anchor-free mechanism named the Recurrent Bidirectional Feature Pyramid Detector (RBFPDet). This model treats helmet-wearing detection as a task of robust semantic feature point detection, significantly improving the accuracy of helmet detection and enabling real-time detection in complex backgrounds. Additionally, [15] introduced an improved safety helmet detection algorithm based on YOLOv8, YOLOv8n-SLIM-CA, which integrates a Coordinate Attention (CA) mechanism and a lightweight neck structure, Slim-Neck, reducing model complexity while maintaining accuracy. Bao et al. [16] integrated the C2F (a faster version of the CSP Bottleneck with two convolutions) module and the FasterNet with EMA (FE) module into the YOLOv8 network architecture, creating a new attention mechanism module named C2F-FE. This module enhances the model's perception of safety helmet targets by fusing features from different levels and incorporating attention mechanisms, simultaneously reducing computational expenses.

III. YOLOv8n-ASF-DH

A. YOLOv8 MODEL

The YOLOv8 model [4], [5], proposed by the Ultralytics team in 2023 and illustrated in Figure 1 [15], encompasses a backbone network, a neck network, and a detection head. The backbone is primarily responsible for extracting high-level semantic features from input images. It employs a cross-stage partial network structure to reduce computational load and enhance gradient flow, incorporating a spatial pyramid pooling module for improved spatial feature extraction. Positioned between the backbone and the head, the neck further integrates and adjusts features extracted by the backbone, facilitating the fusion of different levels of features and enhancing the network's perception of targets. The neck network eliminates the convolutional structure in the upsampling phase of the PAN-FPN from YOLOv5, opting for downsampling followed by upsampling. The C3 module is replaced with the C2f module, lightening the model while improving adaptability to targets of varying sizes and shapes. The detection head, responsible for target localization and classification, generates the final detection outcomes. YOLOv8 transitions from a coupled head structure to the current mainstream decoupled head structure, separating the classification and detection heads and moving from anchor-based to anchor-free.

This paper opts for the YOLOv8n network model as the foundational framework. It enhances the YOLOv8n network architecture to better adapt it to the safety helmet detection dataset and improve the network's detection performance, especially in detecting small targets.

B. TRIPLET ATTENTION MODULE

The human visual system inspires the concept of attention mechanisms. It posits that neural networks should focus on the most relevant or significant aspects when processing complex input information. Integrating attention

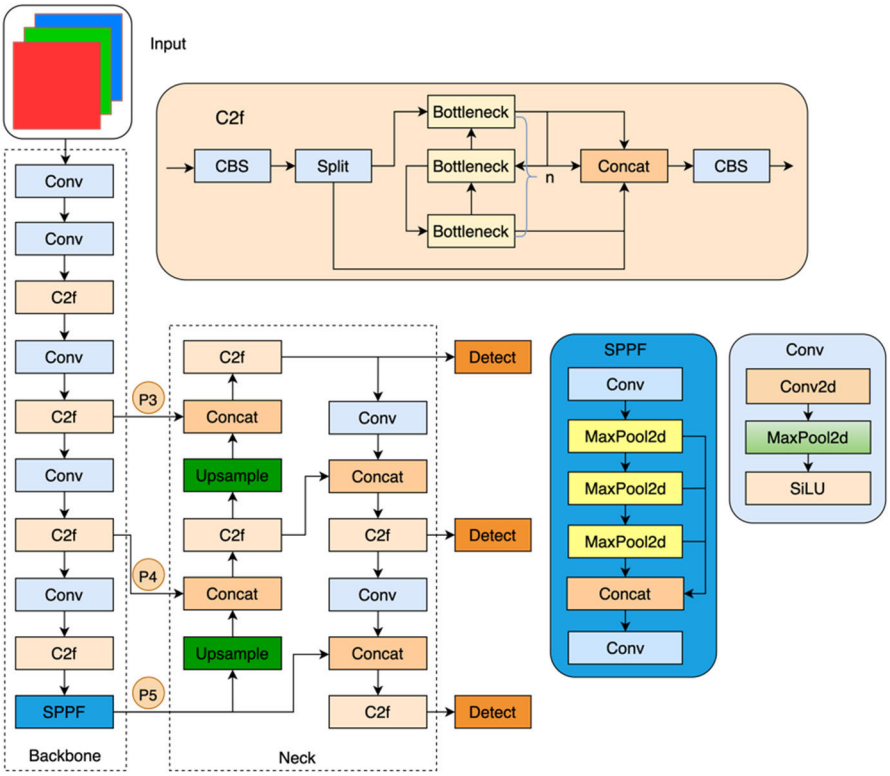


FIGURE 1. The architecture of the YOLOv8n [15].

mechanisms significantly enhances model performance and generalization capabilities [20]. Unlike conventional attention network structures such as SENet [21] and CBAM [22], which require a certain number of learnable parameters to establish inter-channel dependencies, the Triplet Attention module introduces a novel method of calculating attention weights through a three-branch structure that captures cross-dimensional interactions [20]. Triplet Attention establishes inter-dimensional dependencies for input tensors through rotation operations and residual transformations, encoding inter-channel and spatial information with negligible computational overhead. As depicted in Figure 2, Triplet Attention comprises three parallel branches that capture cross-dimensional interactions between channel C and spatial dimensions H or W, respectively. Like that in CBAM, the final branch is dedicated to constructing spatial attention. The outputs of these three branches are aggregated using an averaging method to form a unified attention expression. The implementation of the three branches for a given input dimension of $C \times H \times W$ is as follows:

The first branch focuses on channel dimension C and spatial dimension H interactions to capture relevant features. This process begins by rotating the input feature map 90 degrees counterclockwise along the H axis, transforming it into a $W \times H \times C$ shape. Subsequently, a Z-Pool operation is applied along the W dimension, combining average pooling and max pooling techniques to preserve rich feature representations while reducing data depth, resulting in a $2 \times H \times C$ output feature. The features are then processed through a 7×7 convolutional layer and a batch normalization (BN) layer, generating corresponding attention weights via a sigmoid activation layer. Finally, these attention weights are applied to the original feature map through a residual transformation and rotated 90 degrees clockwise along the H axis, producing a feature map of dimensions $C \times H \times W$. The Z-Pool operation is defined as:

$$Z - Pool(x) = [MaxPool_{od}(x), AvgPool_{od}(x)] \quad (1)$$

where $\text{MaxPool}_{\text{od}}$ refers to max pooling, and $\text{AvgPool}_{\text{od}}$ refers to average pooling.

The second branch aims to capture features through interactions between channel C and spatial dimension W. This branch starts by rotating the input feature map 90 degrees counterclockwise along the W axis, converting it into an $H \times C \times W$ shape. A Z-Pool operation is then performed on the H dimension, resulting in a $2 \times C \times W$ feature map. These features are processed through a 7×7 convolution and BN layer, activated by an activation function. Finally, the attention weights are multiplied back into the original feature map through a residual transformation and rotation operation, restoring it to a $C \times H \times W$ dimension.

The third branch is responsible for computing spatial attention, beginning with processing the input feature map through a channel pool and converting it into a $2 \times H \times W$ shape.

The features are then processed through 7 × 7 convolutional and BN layers, generating spatial attention weights through an activation function. Ultimately, these attention weights are applied to the original feature map through a residual transformation and rotation operation, restoring it to a C × H × W dimension.

The calculation of attention weights in each branch involves several steps:

Input Permutation: For the first branch, the input tensor is permuted to bring dimensions C and W together, resulting in a tensor of shape (C × W × H).

Z-Pooling: Z-pooling aggregates the values along the permuted dimensions, effectively reducing the tensor's dimensionality while preserving the essential features.

Convolutional Operation: The pooled tensor is processed by a convolutional layer with a kernel size k × k, capturing intricate feature interactions within the permuted dimensions.

Sigmoid Activation: The convolutional output is passed through a sigmoid activation function, generating attention weights that range between 0 and 1. These weights highlight the importance of each element within the tensor.

Re-permutation and Application: The attention weights are returned to the input tensor's original dimension alignment. They are then applied to the original tensor, modulating its values according to the calculated attention.

Averaging: The final attention map is obtained by averaging the outputs from the three branches, ensuring that combined attention comprehensively captures interactions across all dimensions.

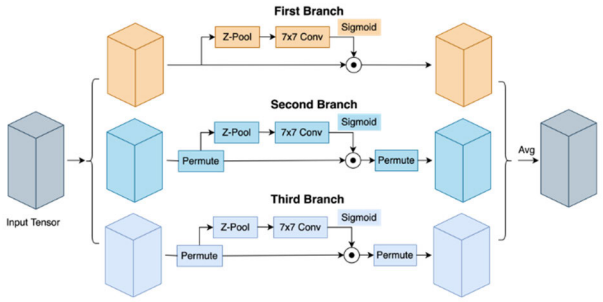


FIGURE 2. The structure of the triplet attention module.

Given the small pixel size of densely worn helmets in images, feature extraction by the backbone network is challenging. Embedding the Triplet Attention module within the backbone network focuses the model on the features of densely packed small targets during the feature extraction phase, enhancing the detection accuracy of small objects. This method effectively enhances the model's capability to capture spatial and channel information through meticulous adjustment and combinations of different operations, thereby enriching and accurately representing features.

To validate the impact of the Triplet Attention module on the performance of safety helmet detection, we conducted experiments comparing the performance of YOLOv8n models integrated with different attention mechanisms on the safety helmet target detection. As shown in Figure 3, the experimental results assess the enhancement capabilities of SE, CBAM, and Triplet Attention mechanisms on feature representation and their impact on model focus. The results, visualized through heatmaps, facilitate qualitative analysis. Through heatmaps of multiple images, we observed that, compared to SE and CBAM attention mechanisms, the Triplet Attention module concentrates more on the location of workers' safety helmets, further highlighting the Triplet Attention module's exceptional capability in feature utilization.

We experimented with various combinations of attention module placements to determine the optimal integration of the Triplet Attention module into YOLOv8. The metrics such as mean Average Precision, precision, and recall are comprehensively explained in Section IV (Experiments), Subsection B (Evaluation Criteria). The baseline model, YOLOv8n, without the Triplet module, is shown in Table 1. The Evaluation Criteria section describes FLOPs, parameters, precision, recall, mAP 0.5, and mAP 0.5:0.95. Adding the Triplet Attention Module does not alter the FLOPs, indicating that this module does not increase the computational complexity. The number of parameters is also roughly the same, indicating that the model size is unaffected. Overall, including the Triplet module enhances model performance, mainly when applied to deeper layers of the model (such as the output of P5). This may be due to the deeper features containing more abstract and richer information, with the Triplet module aiding the model in better capturing these details. The improvements across various metrics suggest that the Triplet module is an effective enhancement that boosts the model's target detection performance. Integrating the Triplet Attention module into the ninth layer of the backbone network (the output of P5) resulted in significant improvements in precision (P) (+1.8%), recall (R) (+1.2%), mAP 0.5 (+0.9%), and mAP 0.5:0.95 (+2.6%) with minimal parameter increase (+5400 B).

Integrating the Triplet Attention module into other positions might reduce the model's detection effectiveness for two reasons: First, increasing the number of attention modules could lead to data redundancy, thereby affecting detection outcomes. Second, the lower layers of the backbone network capture simple, local features, while the higher layers capture more complex, global features. In this context, placing the attention module in higher layers allows the network to focus more on the critical parts of these complex features.

Through experimental comparison, we decided to add the Triplet Attention module to the ninth layer of the backbone network (the output of P5, after C2f), as shown in Figure 4. The figure illustrates the enhanced backbone network structure with Triplet Attention. P2-P5 represents different output levels of the Feature Pyramid Network (FPN), each with varying spatial resolutions suitable for detecting objects of different sizes. P2, with the highest resolution, is ideal for detecting small objects; P3, with a medium resolution, balances the detection of small and medium-sized objects; P4, with a lower resolution, facilitates the detection of medium- to large objects; and P5, with the lowest resolution,

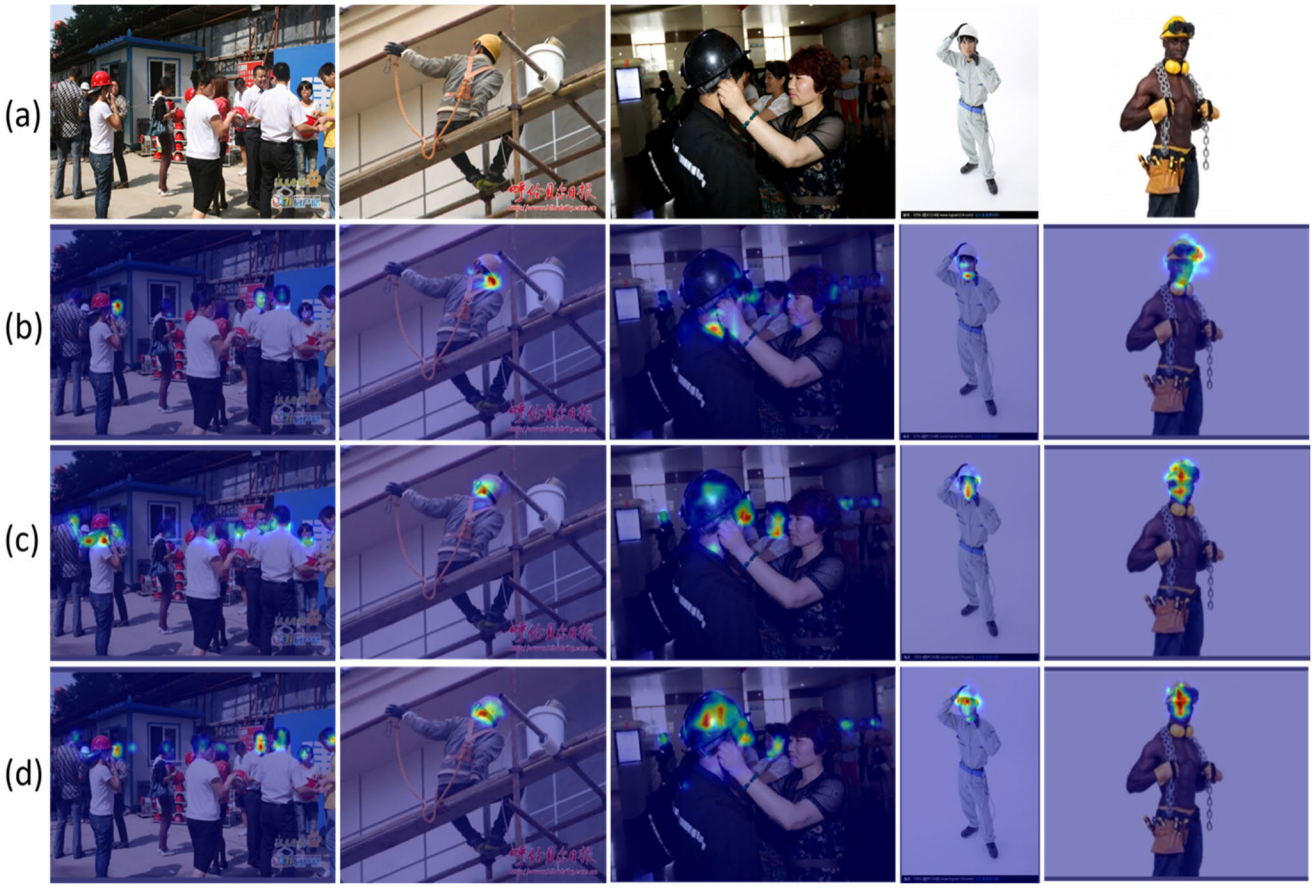


FIGURE 3. The heatmap results of different attention modules. (a) Original image; (b) YOLOv8n + SE heatmap; (c) YOLOv8n + CBAM heatmap; (d) YOLOv8n + Triplet heatmap.

TABLE 1. Comparison of results with a triplet attention module added at different positions.

Addition Locations	FLOPs (GB)	Parameters (MB)	P(%)	R(%)	mAP 0.5(%)	mAP 0.5:0.95(%)
YOLOv8n (No Triplet module)	8.2	3.006233	90.9	86.9	93.0	60.7
Layer 3 (output of P2)	8.2	3.011633	91.6	86.1	92.8	59.9
Layer 5 (output of P3)	8.2	3.011633	91.1	86.6	93.0	60.7
Layer 7 (output of P4)	8.2	3.011633	91.5	87.4	93.4	61.1
Layer 9 (output of P5)	8.2	3.011633	92.7	88.1	93.9	63.3
Layer 3 and Layer 6 (outputs of P2 and P3)	8.2	3.011833	91.8	86.4	93.0	60.7
Layer 3 and Layer 8 (outputs of P2 and P4)	8.2	3.011833	91.6	86.9	93.2	61.0
Layer 3 and Layer 10 (outputs of P2 and P5)	8.2	3.011833	92.0	87.0	93.4	61.4
Layer 5 and Layer 8 (outputs of P3 and P4)	8.2	3.011833	91.9	86.5	93.0	60.7
Layer 5 and Layer 10 (outputs of P3 and P5)	8.2	3.011833	92.7	87.0	93.3	61.4
Layer 7 and Layer 10 (outputs of P4 and P5)	8.2	3.011833	92.4	87.0	93.4	61.1

is primarily used for detecting large objects. These layers enhance the model's ability to detect objects of various sizes by fusing features across different scales.

C. ASF

This paper introduces the ASF structure [23] into the neck layer to better capture information on small targets. ASF, a framework combining spatial and multi-scale features in object detection and segmentation, allows for the fusion of output features extracted from the backbone network's P3, P4, and P5 levels. This feature fusion network architecture, composed of two main component networks, provides complementary information for small object segmentation: the SSFF module combines global or high-level semantic information from images of multiple scales; the TFE module captures local fine details of small target objects. Thus,

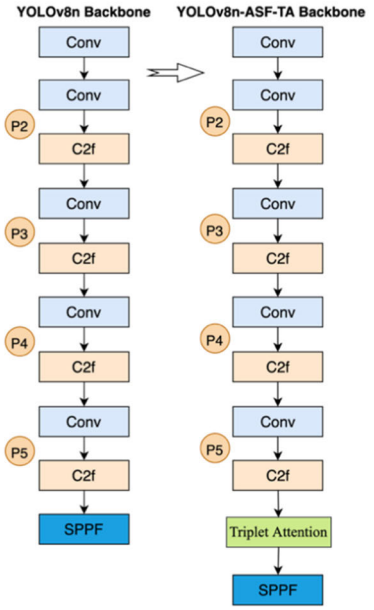


FIGURE 4. Backbone network enhanced with triplet attention.

combining local and global feature information can produce more accurate segmentation maps.

To address the multi-scale issue of safety helmets, existing literature employs feature pyramid structures for feature fusion, typically using summation or concatenation to merge pyramid features [24]. However, various feature pyramid networks must effectively utilize the correlations between all pyramid feature maps. The SSFF scale sequence feature fusion combines better deep feature maps' high-dimensional information with shallow feature maps' detailed information, where image sizes change during the downsampling process, but scale-invariant features remain unchanged. The scale space is constructed along the image's axis, representing not just a single scale but a range of possible scales an object can have. Although a blurred image may lose detail, the structural features of the image can be preserved. The scaled images input into SSFF can be obtained from Equations 2 and 3:

$$F_\sigma(w,h) = G_\sigma(w,h) \times f(w,h) \tag{2}$$

$$G_\sigma(w,h) = \frac{1}{2\pi\sigma^2} e^{-(w^2+h^2)/2\sigma^2} \tag{3}$$

where $f(w, h)$ denotes 2D input images of width w and height h. $F\sigma(w, h)$ is produced by applying a smoothing process through a sequence of convolutions with a 2D Gaussian filter $G\sigma(w, h)$. Here, σ represents the scale parameter that dictates the standard deviation of the 2D Gaussian filter employed in the convolution process.

As shown in Figure 5, the SSFF module enhances feature expression by combining feature maps from different network depths and using 3D convolution to extract cross-level spatial features. These are further processed through BN and SiLU to optimize and introduce non-linearity, followed by a squeeze operation to reduce dimensions.

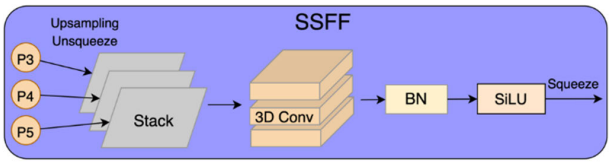


FIGURE 5. Structure of the SSFF module.

The TFE module enhances the detection of densely worn helmets among small objects by concatenating features of different sizes (large, medium, and small) in the spatial dimension to capture detailed information on small targets. The information from the TFE module is then integrated into each feature branch through the PANet structure [24] and combined with the multi-scale information from the SSFF module into the P3 branch. As shown in Equation 4, three feature maps of large, medium, and small sizes with the same dimensions are convolved once and then spliced in the channel dimension.

$$F_{TFE} = \text{Concat}(F_l, F_m, F_s) \tag{4}$$

where F_{TFE} represents the feature maps outputted by the TFE module. F_l, F_m, and F_s correspond to large, medium, and small feature maps, respectively.

As indicated in Table 2, compared to the YOLOv8n model, the YOLOv8n+ASF model has slightly higher FLOPs and more parameters but also significantly improves precision, recall, and mAP under two IoU thresholds.

TABLE 2. Impact of adding the ASF module on model performance.

Model	FLOPs (GB)	Parameters (MB)	P(%)	R(%)	mAP 0.5(%)	mAP 0.5:0.95(%)
YOLOv8n	8.2	3.006233	90.9	86.9	93.0	60.7
YOLOv8n +ASF	8.6	3.051561	93.1	88.5	94.4	63.3

D. DYHEAD

The YOLOv8 detection head adopts a state-of-the-art decoupled head design, transitioning from anchor-based to anchor-free, significantly enhancing the model's robustness and adaptability. This advancement enables YOLOv8 to process target detection tasks in complex scenarios more efficiently, offering a more reliable solution for practical application scenarios. Despite these improvements, the model still needs a unified approach to addressing detection challenges.

This study integrates attention mechanisms into the target detection module, transforming the detection head into a unified dynamic detection head, termed Dynamic Head (DyHead). This innovation facilitates the synergistic operation of multiple self-attention mechanisms across feature layers, promoting scale perception, spatial awareness, and task sensitivity enhancements. This significantly bolsters the representative capability of the target detection module without incurring additional computational costs [26]. By ingeniously incorporating attention mechanisms across critical dimensions of the feature tensor, this design achieves

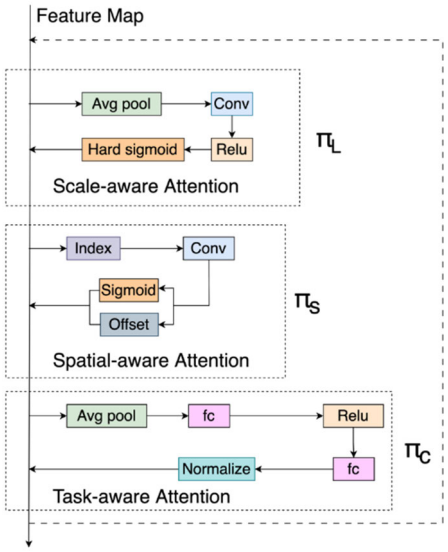


FIGURE 6. Structure of the DyHead detection head network.

a unified and dynamic adjustment of these three critical perceptual abilities.

Images are initially processed through the backbone and neck networks, yielding three-scale feature maps. As illustrated in Figure 6, these feature maps are processed by a scale-aware attention module that handles hierarchical features L, merging features of various scales to enhance expressiveness and sensitivity to scale variations of foreground objects. Subsequently, spatial features S are refined by a spatial awareness attention module, improving the expression of features at different positions within the feature map, thus focusing more on the spatial locations of foreground objects. Additionally, channel feature C is enhanced by a task-aware attention module, further elevating the expressiveness of features. The detection head network handles these meticulously processed features, producing the final detection outcomes. DyHead endows the detection head with enhanced representational power, improving the algorithm's effectiveness in detecting small objects. Given a three-dimensional feature tensor $F \in R^{L.\times S.\times C}$, the attention function is computed as follows:

$$W(F)\& = \pi_C(\pi_S(\pi_L(F) \cdot F) \cdot F) \cdot F \tag{5}$$

where $\pi_L(\cdot)$, $\pi_S(\cdot)$, $\pi_C(\cdot)$ are three different attention functions applied to dimensions L, S, and C, respectively.

Exploring the impact of stacking varying numbers of DyHead blocks on model performance and computational costs, the baseline model utilized is YOLOv8n, with experiments conducted on stacking 0 (no addition, serving as the baseline YOLOv8n model) to 1, 2, 4, 6, 8, 10, and 12 DyHead blocks. The results, as shown in Table 3, lead to the following conclusions:

1. With increased DyHead layers, FLOPs and parameter counts rise, indicating a more complex model with higher computational demands.

2. Precision (P) and recall (R) improve with increasing DyHead layers, indicating performance enhancements.

3. mAP 0.5 and mAP 0.5:0.95 generally increase with more DyHead layers but not linearly, suggesting diminishing returns in performance improvements. Performance gains begin to plateau beyond a certain number of DyHead layers.

These findings suggest that adding more DyHead layers can enhance the detection performance of the YOLOv8 model, albeit at the cost of increased computation. Thus, this paper finds a balance between performance and computational efficiency, considering algorithm accuracy, computational costs, and model parameter volume, opting to integrate four DyHead blocks into the algorithm. At this juncture, the model's mAP 0.5 increased by 0.868% and mAP 0.5:0.95 by 2.445%, with a parameter increase of 1.971M and only a 3.1GB increase in FLOPs.

E. FOCAL-EIOU

YOLOv8 employs CIoU Loss [27] as its bounding box loss function. CIoU Loss, by incorporating overlap area, center distance, and aspect ratio into its calculations, enhances model training stability and convergence speed. However, CIoU Loss needs to accurately reflect the actual disparity between the dimensions of the bounding box and its confidence, potentially leading to imprecise regression predictions. Moreover, considering that the safety helmet dataset contains images of varying quality across different scenes, the diversity in data collection environments and locations includes low-quality sample images. Geometric measures such as distance and aspect ratio can exacerbate penalties on low-quality samples, thus degrading the model's generalization performance. The CIoU used in YOLOv8 faces this issue: when encountering low-quality samples, the loss function should weaken the penalty on geometric measures when the anchor box and target box align well.

This paper introduces the Focal-EIoU [30] loss function to address the issues with CIoU loss. Focal-EIoU, a novel loss function formed by integrating FocalL1 loss and Efficient IoU (EIoU), introduces EIoU on top of CIoU's penalty terms, separately calculating the lengths and widths of the actual and predicted boxes. EIoU includes overlap loss, center distance loss, and aspect ratio loss, leading to faster convergence and more stable bounding box regression. This efficient IoU variant balances the contributions of high-quality and low-quality samples in the loss calculation, helping the model focus on learning from challenging samples and improving its overall performance. The EIoU is calculated as follows:

$$E_{\text{IoU}} = \frac{\rho^2(b, b^{\text{gt}})}{c^2} + \frac{\rho^2(w, w^{\text{gt}})}{c_{\text{w}}^2} + \frac{\rho^2(h, h^{\text{gt}})}{c_{\text{h}}^2} \tag{6}$$

where $\rho(w, w^{\text{gt}})$ represents the difference in width between the predicted bounding box and the ground truth bounding box; $\rho(h, h^{\text{gt}})$ represents the difference in height between the predicted bounding box and the ground truth bounding box; cw is the width of the smallest bounding box that encloses

TABLE 3. Impact of the number of DyHead layers on model performance.

the Number of DyHead Blocks	FLOPs (GB)	Parameters (MB)	P(%)	R(%)	mAP 0.5(%)	mAP 0.5:0.95(%)
0 (YOLOv8n)	8.2	3.006233	90.9	86.9	93.0	60.7
1	9.7	3.491237	92.7	87.8	93.8	62.7
2	10.3	3.986657	92.7	88.2	93.9	63.1
4	11.3	4.977497	93.4	88.1	94.3	63.1
6	12.4	5.968337	93.5	88.5	94.4	63.7
8	13.5	6.959177	93.5	88.5	94.5	63.6
10	14.5	7.950017	93.6	88.6	94.6	64.1
12	15.6	8.940857	93.3	89.2	94.8	64.1

both boxes; ch is the height of the smallest bounding box that encloses both boxes.

The final EIoU loss function can be written as Equation 7.

$$L_{EIoU} = L_{IoU} + L_{dis} + L_{asp} \tag{7}$$

where L_{IoU}, L_{dis}, and L_{asp} represent IoU loss, distance loss, and aspect ratio loss, respectively. In a sample image, the number of anchor boxes with minor regression errors is far fewer than those with significant errors. Poor-quality anchor boxes generate more significant gradients, directly affecting the training outcome. Therefore, by adding focal loss to EIoU loss, high-quality anchor boxes are distinguished from low-quality ones, as shown in the formula:

$$L_{Focal-EIoU} = IoU^{\gamma} L_{EIoU} \tag{8}$$

where γ is a hyperparameter used to control the curvature of the curve.

EIoU loss directly minimizes the difference in width and height between the target and anchor boxes, leading to faster convergence and better localization results. Reweighting EIoU loss with the value of IoU to focus on high-quality target boxes further enhances its performance.

To investigate the impact of different loss functions on the YOLOv8n algorithm, we conducted experiments with commonly utilized loss functions, including CIoU, GIoU, DIoU, EIoU, and Focal-EIoU. The training accuracy achieved by substituting the original CIoU loss in the YOLOv8n algorithm with these varied loss functions is presented in Table 4. The table reveals that different loss functions have distinct effects on the model's performance. Compared to other loss functions, Focal-EIoU offers the most significant improvement in the model's detection capabilities, indicating that the Focal-EIoU loss function is more suitable for object detection in the safety helmet dataset. The focal-EIoU loss function balances the contributions of high-quality and low-quality samples to the loss, i.e., it enhances the contribution of high-quality (high IoU) samples while suppressing the contribution of low-quality (low IoU) samples. Therefore, the loss function selected for this study is Focal-EIoU.

TABLE 4. Comparison of results with different loss functions.

Loss Function	P(%)	R(%)	mAP 0.5(%)	mAP 0.5:0.95(%)
CIoU [27]	90.9	86.9	93.0	60.7
GIoU [28]	92.6	86.5	92.3	60.7
DIoU [29]	91.9	86.7	92.5	60.8
EIoU [30]	92.3	87.9	93.6	62.4
Focal-EIoU [30]	92.9	88.1	93.9	62.9

F. ENHANCED YOLOv8 MODEL

The proposed enhanced YOLOv8 network architecture, YOLOv8n-ASF-DH, is shown in Figure 7 and incorporates several improvements over the original YOLOv8 framework. Integrating the Triplet Attention mechanism into the backbone layer optimizes the model's focus on densely packed small targets during the feature extraction phase, significantly improving detection accuracy for such targets. The introduction of the ASF structure in the neck layer merges spatial and multi-scale features, efficiently fusing output features extracted from the backbone network's P3, P4, and P5 levels. This ASF structure markedly enhances the model's recognition of small target details. Employing DyHead in the detection head enables the synergistic operation of multiple self-attention mechanisms across feature layers, effectively promoting the model's scale awareness, spatial perception, and task sensitivity. This advanced mechanism significantly strengthens the representational power of the target detection module, providing robust support for the accurate identification and localization of targets.

Furthermore, adopting the Focal-EIoU loss function achieves a balanced adjustment of contributions from high-quality and low-quality samples within the overall loss calculation, increasing the weight of high-quality (high IoU) samples while reducing the impact of low-quality (low IoU) samples. This approach effectively enhances the model's focus on high-quality samples, optimizing the learning process and model performance. Through these various enhancements, the YOLOv8n-ASF-DH model achieves improved target detection for safety helmet-wearing, especially in detecting small targets.

This comprehensive approach, integrating advanced attention mechanisms, structural optimizations, and a refined loss function, culminates in a significantly improved version of the YOLOv8 model. The YOLOv8n-ASF-DH model demonstrates superior performance in detecting safety helmets, particularly among small objects, and showcases the potential for broad applicability across different object detection

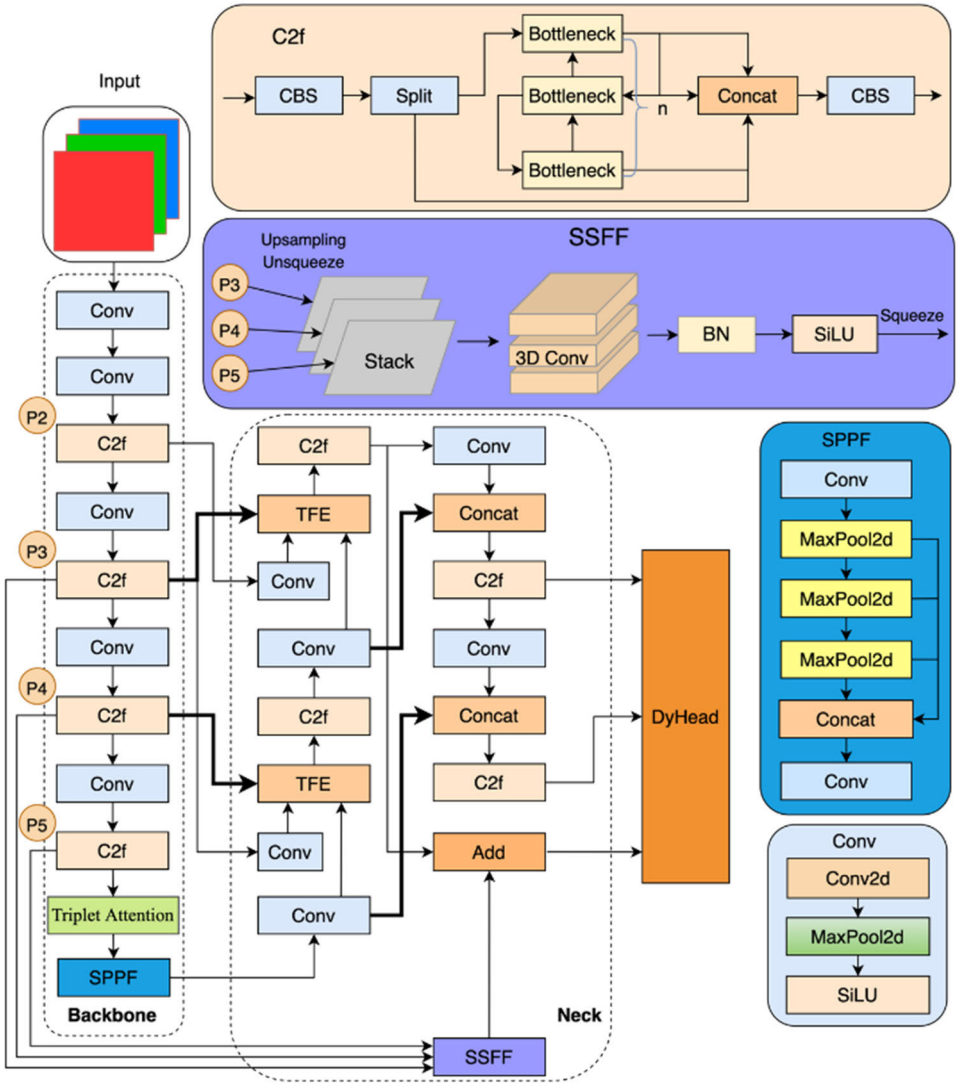


FIGURE 7. Structure of the YOLOv8n-ASF-DH network.

tasks. The model's advancements underscore the importance of continuous innovation in deep learning architectures and loss functions to meet the evolving challenges of real-world applications.

IV. EXPERIMENTS

A. DATASETS AND EXPERIMENTAL ENVIRONMENT

The efficacy of the proposed YOLOv8n-ASF-DH model was assessed with the publicly available SHWD [31]. This dataset is designed to detect safety helmet usage and identify human heads, comprising 7,581 images, including 9,044 individuals wearing safety helmets (positive samples) and 111,514 instances of head objects without helmets (negative samples). The dataset is organized according to YOLO standards into two categories: individuals without safety helmets and those wearing them. It also provides bounding box coordinates for target localization. The SHWD dataset is accessible at https://github.com/njvisionpower/Safety-Helmet-Wearing-Dataset. It was randomly divided into training, validation, and test sets in a 7:2:1 ratio.

During model training, data augmentation operations are applied to the dataset. The online data augmentation techniques include Mosaic augmentation, Mixup augmentation, random perspective perturbation, and HSV color augmentation.

The experiments used an NVIDIA RTX 3080 Ti (12G) GPU with PyTorch 2.10 and Python 3.8 dependencies. We utilized the initial weights of the pre-trained YOLOv8 model. The input image resolution was set to 640 × 640 pixels. The batch size was 16, and the training times were 200 epochs. Stochastic Gradient Descent (SGD) was employed as the optimization algorithm, with hyperparameters set to a momentum of 0.937, an initial learning rate of 0.001, and a weight decay of 0.0005.

B. EVALUATION CRITERIA

Object detection evaluation metrics are introduced to facilitate comparison among different algorithm models.

In Table 5, true positive (TP) is the case where the actual sample is positive and predicted to be positive. If a person is indeed wearing a safety helmet and the model correctly identifies that this person is wearing a safety helmet, this identification result is referred to as TP. False negative (FN) is the case where the actual sample is positive but is predicted to be negative, also known as a false negative. If a person is wearing a safety helmet, but the model fails to recognize that this person is wearing a safety helmet, this omission is an FN. False positive (FP) is the case where the actual sample is negative but is predicted as positive, also known as a false positive. If the model mistakenly marks a non-helmet-wearing area as wearing a safety helmet, this error is an FP. True negative (TN) is the case where the actual sample is negative and predicted as negative, also known as a true negative. If an area does not have a safety helmet and the model correctly identifies that the area does not have any targets to recognize, this situation is a TN.

TABLE 5. Confusion matrix.

Confusion Matrix		Actual Condition	
		Positive	Negative
Predicted Condition	True	TP	TN
	False	FP	FN

1. Intersection over Union (IOU) is a metric used in object detection tasks. It is calculated as the ratio of the intersection area between the predicted bounding box and the ground truth bounding box to the union area of the two bounding boxes.

$$IoU = \frac{\mathrm{A} \cap \mathrm{B}}{\mathrm{A} \cup \mathrm{B}} \tag{9}$$

where IoU ranges from 0 to 1, where A represents the predicted bounding box, and B represents the ground truth bounding box.

2. Precision (P) quantifies the fraction of true positive detections about the sum of true positive and false positive detections. It is indicative of the accuracy of the positive class predictions.

$$Precision = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{10}$$

3. Recall (R) measures the proportion of actual positives correctly identified by the detection model, thus reflecting the model's ability to detect relevant instances.

$$Recall = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{11}$$

4. Average Precision (A_P) provides a single-figure quality measure across recall levels. It captures the precision-recall trade-off by computing the average precision value for recall values over the interval [0, 1].

$$A_P = \int_0^1 P(r)\,\mathrm{d}r. \tag{12}$$

where p(r) represents the precision at a given recall r.

5. mAP is a composite metric that calculates the mean of the average precision for each class across varying thresholds of IoU, offering a consolidated measure of the model's precision across multiple object classes.

$$\mathrm{mAP} = \frac{\sum_{i=1}^{N} A_{P,i}}{\mathrm{N}} \tag{13}$$

where denotes the average precision for class i, and N denotes the total number of classes.

6. Floating Point Operations (FLOPs) are an acronym for Floating Point Operations Per Second, and they are a standard metric for quantifying the computational complexity of an algorithm or a model. Specifically, within neural networks, FLOPs often refer to the number of floating-point operations required to generate a forward pass through the network.

7. Detection speed, measured in frames per second (fps), refers to the number of images that can be processed for detection within one second.

8. In neural networks, parameters refer to the weights and biases learned during training. The total count of these learnable parameters directly indicates the model's capacity, complexity, and memory requirements.

C. MODEL TRAINING

Figure 8 shows the performance comparison of YOLOv8n and its improved model, YOLOv8n-ASF-DH, over 200 training epochs. Performance metrics include mean average precision (mAP) and total loss, which change as the training epochs increase.

Figure 8 (a) describes the mAP 0.5 (i.e., mean precision when the IoU threshold is 0.5) of the two models across different training epochs. mAP is a standard metric for measuring the accuracy of object detection models, with higher mAP values generally indicating better model performance. The curve shows that YOLOv8n-ASF-DH (red line) has higher mAP values than the original YOLOv8n model (black line) for most of the training epochs. This indicates that the model integrated with the SLIM-CA mechanism has higher accuracy in object detection tasks.

Figure 8 (b) displays the total loss curves of the two models. In deep learning, the loss function estimates the inconsistency between the model's predictions and the actual values, and a reduction in total loss during training is one indicator of model performance improvement. The graph shows that the loss of both models decreases as the training epochs increase, but the loss of YOLOv8n-ASF-DH decreases more significantly and remains lower than that of YOLOv8n.

Figure 8 indicates that performance improvements in object detection tasks have been achieved by enhancing the YOLOv8n model with four improvement modules. Specifically, the improved model has increased accuracy and has shown faster convergence and lower loss during the learning process.

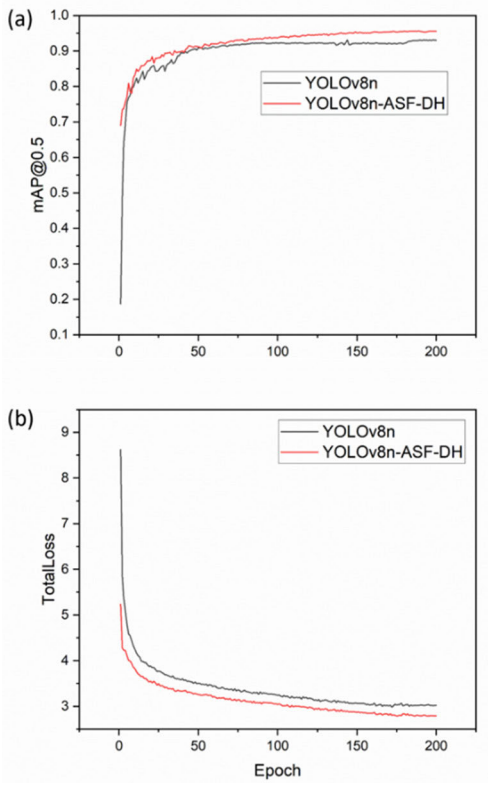


FIGURE 8. Comparison of mAP and loss between YOLOv8n and YOLOv8n-ASF-DH. (a) mAP(%). (b) loss.

D. ABLATION STUDY

We conducted ablation experiments based on the proposed method to measure the impact of different modules on the performance of the safety helmet detection model and verify the optimization effects of each module in the network. As shown in Table 6, the experimental results use YOLOv8n as the baseline model and consider four modules: the Triplet Attention module, ASF structure, DyHead, and the Focal-EIoU loss function. Each row in the table represents a combination experiment of different modules, with selected modules indicated by a checkmark. The results from the table show:

1. All modules improve model performance when used individually, but the effects are better when combined.

2. The last row, which uses all four modules (our model YOLOv8n-ASF-DH), shows the highest performance metrics, with precision, recall, mAP 0.5, and mAP 0.5:0.95 increasing by 3.1%, 3.9%, 2.6%, and 6.1%, respectively, compared to the YOLOv8n model. This indicates that these modules work best together, providing the best detection performance. With all components used, mAP 0.5 and mAP 0.5:0.95 reached 95.6% and 66.8%, respectively, indicating the high robustness of the model under different IoU thresholds.

3. Performance metrics are the lowest without any proposed components, indicating that each component positively enhances the model's detection capability.

4. The recall rate (R) fluctuates across all experimental setups but reaches its highest value when all components are used together, indicating a complementary effect between components.

Figure 9 presents the Precision-Recall Curve, a chart used to evaluate the performance of classification models. The curve shows the detection results for the head, helmet, and a combination of all categories. Points on the curve represent precision values at different recall levels. For the head category, the model achieved a maximum precision of 0.944. For the helmet category, the model reached a maximum precision of 0.968. Combining all categories, the model achieved a mAP 0.5 (mean precision at an IoU threshold of 0.5) of 0.956. From the results, it can be inferred that the YOLOv8n-ASF-DH model is accurate and reliable in detecting both head and helmet-wearing categories.

E. EXPERIMENTAL COMPARISON WITH OTHER METHODS

This section compares the enhanced model introduced in this research against other prevalent detection models within the SHWD dataset. The experimental outcomes are comprehensively delineated in Table 7. It can be observed from the table that:

1. YOLOv8n is the fastest, completing inference in just 2.1ms; however, it does not achieve the best performance regarding mAP 0.5 and mAP 0.5:0.95.

2. Improved YOLOv8 and YOLO-M boast comparably high mAP 0.5 scores of 94.4% and 93.9%, respectively, yet Improved YOLOv8 exhibits a marginally higher mAP 0.5:0.95 indicating its superior performance at stricter IoU thresholds.

3. YOLOv7-w6 also shows commendable results in mAP 0.5:0.95, attaining 58.8%, while maintaining a relatively rapid processing speed of 18.1ms.

4. YOLOv8n-ASF-DH scores the highest in mAP 0.5:0.95 at 66.8%, although its speed is the slowest among all listed algorithms at 12.1ms.

Among all the evaluated models, the algorithm proposed in this paper is the most efficacious, registering exceedingly high levels on two pivotal metrics. It secured an mAP 0.5 of 95.6% and an mAP 0.5:0.95 of 66.8%. The YOLOv8n-ASF-DH achieves the optimal mAP values without substantially increasing the computational load and the number of parameters.

In summary, Table 7 unequivocally highlights the pronounced benefits of the YOLOv8n-ASF-DH in enhancing accuracy and computational efficiency.

F. CASE ANALYSIS

The SHWD dataset was detected and evaluated using the YOLOv8n model and the improved YOLOv8n-ASF-DH model, with the results shown in Figure 10. The first row uses the YOLOv8 model, and the second uses the YOLOv8n-ASF-DH model. For each image, different frames of different colors represent different detection results.

Green frame (TP): The model correctly identified a person wearing a safety helmet.

TABLE 6. Comparative analysis of ablation experiment.

Modules				P (%)	R (%)	mAP 0.5(%)	mAP 0.5:0.95(%)
Triplet Attention	ASF	DyHead	Focal-EIoU				
				90.9	86.9	93.0	60.7
✓				92.7	88.1	93.9	63.3
	✓			93.1	88.5	94.4	63.3
		✓		93.4	88.1	94.3	63.1
			✓	92.9	88.1	93.9	62.9
✓	✓			93.8	89.3	94.9	64.6
✓		✓		94.0	88.5	94.8	64.0
✓			✓	93.4	88.4	94.5	63.1
	✓	✓		93.9	89.3	95.1	65.0
	✓		✓	93.4	88.9	94.7	64.2
		✓	✓	93.5	88.6	94.4	64.2
✓	✓	✓		93.7	90.5	95.4	65.8
✓	✓	✓	✓	94.0	90.8	95.6	66.8

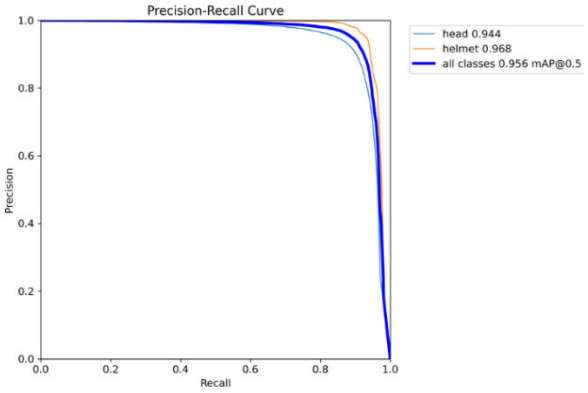


FIGURE 9. Precision-recall curve of YOLOv8n-ASF-DH on the SWHD dataset.

Red frame (FN): This indicates the model failed to recognize a person wearing a safety helmet.

Blue frame (FP): Indicates the model mistakenly marked a person not wearing a safety helmet or mistakenly marked a non-existent safety helmet in the scene.

Based on Figure 10, the detection results are as follows:

(a) Outdoor environment: The YOLOv8n model misidentified a person's arm wearing a safety helmet. This may be due to the overall dim background of the picture, leading to poor model detection performance.

(b) External construction scene: The YOLOv8n model misjudged the arm of an excavator as a target, possibly due to the overly cluttered background.

(c) Construction site entrance scene: The YOLOv8n model misjudged the background as a target, possibly because the background is similar to the target, causing the model to misjudge.

(d) High-altitude outdoor work (grayscale image): The YOLOv8n failed to detect the target wearing a safety helmet on the left, possibly because most of the training process involved color images. In contrast, the test image was grayscale, leading to insufficient model robustness.

(e) In front of a promotional poster: The YOLOv8n model misjudged the poster image as a worker wearing a safety helmet. This may be because the poster image resembles the real target of wearing a safety helmet, leading to misjudgment.

(f) Confusing scene: The YOLOv8n model mis-detected a target holding a safety helmet, where the person not wearing a safety helmet should not be detected as a target.

The YOLOv8n-ASF-DH model demonstrated its outstanding robustness and efficient detection capability in diverse and complex backgrounds. The model maintained a high accuracy rate across various situations and accurately identified targets in various scenes, regardless of their size.

Figure 11 shows the detection effect of small targets wearing safety helmets in different scenes, with the recognition situations as follows:

(a) People walking on the construction site: The YOLOv8n model missed targets wearing safety helmets that were relatively far away. This may be because the color of the missed target's clothing on the left is similar to the color of the safety helmet, possibly leading to model misjudgment; the missed target on the right may be obscured by the person in front, causing the model to misjudge.

(b) Riverside outside the construction site: The YOLOv8n model misjudged distant trees and nearby objects as targets, possibly due to the overly cluttered background.

(c) Worker standing on machinery: The YOLOv8n model misjudged the blue clothing as a target, possibly because the clothing color is the same as the color of the safety helmet, leading to model misjudgment.

(d) Snowy construction site image: The YOLOv8n model misjudged the front wheel of a distant car and nearby clothing as people wearing safety helmets. This may be because their shapes are similar to safety helmets, presenting semi-circular shapes.

(e) Worker renovating the exterior wall of a building: The YOLOv8n model misjudged two smaller targets of safety belts as workers wearing safety helmets. This may be because the color of the safety belts is the same as the safety helmets (yellow), leading to misjudgment.

(f) Workers building scaffolding: Here, targets of various sizes appear, and the YOLOv8n model missed the small target wearing a safety helmet behind the scaffolding. This may

TABLE 7. Comparison of results with other algorithms.

Algorithm	FLOPs (GB)	Parameters (MB)	mAP 0.5(%)	mAP 0.5:0.95(%)	Speed (ms)
YOLOv8n	8.2	3.01	93.0	60.7	2.1
Improved YOLOv8 [15]	11.3	2.74	94.4	61.8	2.3
YOLO-M [17]	-	5.5	93.9	-	-
PG-YOLO [18]	-	0.63	93.3	-	32.9
YOLOv5s-Improved [19]	-	5.06	92.4	59.2	7.0
YOLOX(ECA-Net) [32]			90.5	-	14
MD-YOLO v2 [33]	-	17.5	87.4	-	8
YOLOv5s-DM	-	7.06	90.2	56.5	7.5
YOLOv6s	-	-	90.9	57.9	12.7
YOLOv7-w6	-	69.83	91.7	58.8	18.1
YOLOv8n-ASF-DH (Ours)	11.8	5.017625	95.6	66.8	12.1

be because the scaffolding blocks the area where the worker is wearing a safety helmet or because too many targets of different sizes appear, causing the model to miss.

The YOLOv8n-ASF-DH model demonstrated relatively robust detection performance in different complex scenes, accurately identifying small targets wearing safety helmets in most situations. To reduce the rate of missed detections, the backbone network adds the TA attention mechanism, making the model more focused on all helpful information. Integrating the TA attention mechanism, ASF module, and DyHead detection head into YOLOv8 reduced the model's false and missed detection rates and improved the detection capability for small-scale objects. Compared to the YOLOv8n algorithm, the proposed improved model demonstrates effectiveness and superiority in detecting small targets in various scenes.

Compared to the original YOLOv8n algorithm, the improved model proposed in this study shows clear advantages and improvements in identifying targets of various sizes in all kinds of scenes regarding efficacy and performance.



FIGURE 10. Comparison of detection performance in various complex scenarios between YOLOv8n (first row) and YOLOv8n-ASF-DH (second row). The green, blue, and red boxes denote true positive (TP), false positive (FP), and false negative (FN) predictions, respectively. (a) Outdoor environment; (b) External construction scene; (c) Construction site entrance; (d) High-altitude outdoor work (grayscale image); (e) In front of a promotional poster; (f) Confusing scene.

V. DISCUSSION

A. MODEL PERFORMANCE ANALYSIS

The proposed YOLOv8n-ASF-DH model demonstrates significant performance improvements across multiple evaluation metrics. First, as illustrated by the training curves in Figure 8, the enhanced YOLOv8n-ASF-DH model consistently exhibits higher mAP values than the original YOLOv8n model throughout most of the training epochs. Additionally, the downward trend of the loss function curve indicates that the YOLOv8n-ASF-DH model converges more quickly and maintains a lower total loss value, further validating the effectiveness of the model's optimization strategies.

B. ABLATION STUDY RESULTS

The ablation experiments validate each enhancement module's individual and combined contributions to the model's performance. As shown in Table 6, applying individual modules (such as the Triplet Attention, ASF structure, DyHead, or Focal-EIoU loss function) results in performance improvements. However, the most significant performance gains are observed when combining multiple modules. Specifically, the results in the last row demonstrate that the YOLOv8n-ASF-DH model, which incorporates all proposed modules, achieves the highest performance metrics (Precision, Recall, mAP 0.5, and mAP 0.5:0.95). This indicates the synergy of the combined modules in enhancing the model's detection capabilities.

C. COMPARISON WITH OTHER METHODS

The results presented in Table 7 further validate the superior performance of the YOLOv8n-ASF-DH model. Although the YOLOv8n-ASF-DH model has a slower inference speed (11.8 ms), it achieves the highest mAP 0.5 and mAP 0.5:0.95 values, at 95.6% and 66.8%, respectively. In comparison, other models exhibit a balance of performance and speed, but the YOLOv8n-ASF-DH model stands out regarding detection accuracy under stricter IoU thresholds.

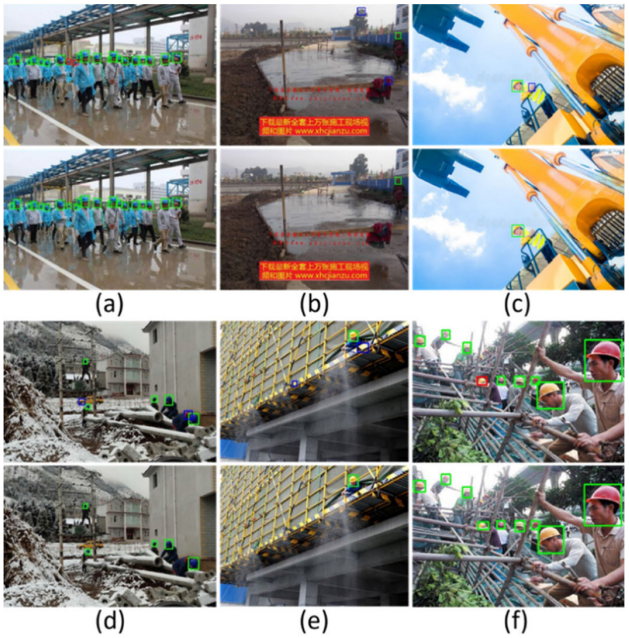


FIGURE 11. Comparison of detection performance for small objects between YOLOv8n (first row) and YOLOv8n-ASF-DH (second row). The green, blue, and red boxes denote true positive (TP), false positive (FP), and false negative (FN) predictions, respectively. (a) People walking at the construction site; (b) Riverside outside the construction site; (c) Workers standing on machinery; (d) Snowy construction site; (e) Workers renovating the exterior wall of a building; (f) Workers building scaffolding.

This underscores the model's applicability in scenarios requiring high precision.

D. CASE ANALYSIS

Figures 10 and 11 illustrate the robustness and efficiency of the YOLOv8n-ASF-DH model in complex scenarios. The model maintains high accuracy in diverse backgrounds, from outdoor environments to high-altitude work sites with significant occlusions. Notably, in small object detection, the enhanced model shows superior recognition capabilities compared to the original YOLOv8n, reducing the rates of false positives and missed detections. Integrating the Triplet Attention mechanism and DyHead detection head enables the model to focus more effectively on relevant information, enhancing its ability to detect small objects.

E. LIMITATIONS AND FUTURE WORK

Despite the outstanding performance of the YOLOv8n-ASF-DH model, its slower inference speed could limit its application in real-time scenarios. Future work should optimize the inference speed without significantly compromising accuracy through model pruning and quantization techniques. Additionally, while this study focuses on safety helmet detection, further research is needed to validate the applicability of the proposed improvements to other object detection tasks.

VI. CONCLUSION

In response to the issues of many small targets, the simultaneous appearance of helmets of different sizes, and complex detection scenes in the task of safety helmet-wearing detection, we propose the YOLOv8n-ASF-DH helmet-wearing detection model. Through a series of comparison and ablation experiments, the following conclusions are drawn:

The proposed model YOLOv8n-ASF-DH, by integrating the Triplet Attention mechanism, introducing the ASF structure and DyHead, and adopting the Focal-EIoU loss function on the YOLOv8n, optimizes the feature extraction and fusion process. It enhances the model's scale, spatial, and task awareness capabilities and balances the loss between high-quality and low-quality samples by adjusting the sample weights in the loss function. These improvements collectively enhance model performance, significantly improving the model's detection capability for various targets, even tiny targets, in complex scenes.

Compared to the baseline YOLOv8n model, with a slight increase in computational and parameter volume, the YOLOv8n-ASF-DH model shows improvements in precision (P), recall (R), mAP 0.5, and mAP 0.5:0.95 by 3.066%, 3.883%, 2.584%, and 6.131%, respectively. This indicates that the improved model significantly enhances the performance of the safety helmet detection task.

However, the model may still produce false and missed detections when the background is complex and noisy. Future work will consider incorporating segmentation algorithms as a pre-processing step for images to reduce the impact of complex backgrounds and noise, thereby improving the model's detection capability.

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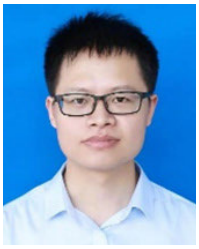
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